

Data-Driven Investment Intelligence

Using NIFTY-50 Market Data



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Enrollment	23117102
Programme	B.Tech Mechanical Engineering — 4Y
Dataset	NIFTY-50 Historical Market Data (Jan 2000 – Apr 2021)
Focus	AI-Powered Decision Support Portfolio Optimization XAI

1. Exploratory Data Analysis (EDA)

The NIFTY-50 dataset spans January 2000 to April 2021 — over 21 years of daily OHLCV data for India's benchmark equity index. After loading and standardising column names (Title Case to lowercase), 49 of 50 stocks were successfully ingested, covering 14 industry sectors and up to 5,306 trading days per ticker.

1.1 Dataset Summary

Attribute	Value
Stocks loaded	49 of 50
Date range	Jan 3 2000 – Apr 30 2021
Sectors represented	14
Max trading days (per stock)	5,306
Average daily volume (RELIANCE)	~14 M shares
Avg pairwise correlation (top-7)	0.257
RELIANCE annualised volatility	38.2%

1.2 Sector Distribution

Financial Services dominates with 9 constituents, followed by Energy (7), Consumer Goods (6), Metals and Automobile (5 each), and IT (5). The remaining sectors — Cement, Pharma, Telecom, Construction, Services, Fertilisers, and Media — contribute 1–3 stocks each.

1.3 Sector Performance (2000–2021)

Sector	Stocks	Avg Total Return	Avg Volatility
Cement & Cement Products	3	15,179%	37.3%
Financial Services	9	2,365%	45.5%
Automobile	5	1,878%	40.8%
Telecom	1	1,110%	40.0%
Consumer Goods	6	377%	36.4%
Metals	5	133%	46.1%
Energy	7	120%	38.4%
IT	5	13%	46.0%
Media & Entertainment	1	-84%	52.6%

Note: Nominal total returns over 21 years; risk-unadjusted.

1.4 Return Characteristics & Fat Tails

Daily returns for RELIANCE exhibit a mean of 0.07% and a standard deviation of 2.40%. The distribution shows extreme fat tails — skewness of -3.64 and excess kurtosis of 82.31 — with a single-day best gain of +21.37% and worst loss of -51.53%. Positive trading days occurred 51.6% of the time. All 49 stocks show excess kurtosis > 0, confirming that normality-based VaR consistently underestimates real tail risk.

Metric	RELIANCE	Implication
Mean daily return	0.07%	Slight positive drift
Std deviation	2.40%	High day-to-day swing
Skewness	-3.64	Left tail (crash risk)
Excess kurtosis	82.31	Extreme outliers likely
Best single day	+21.37%	Post-election/news
Worst single day	-51.53%	Crisis event
% positive days	51.6%	Slight upward bias

1.5 Market-Wide Event Detection

281 trading dates were identified where more than 40% of stocks simultaneously moved more than 3% in a single session. The top events below align precisely with known macro shocks:

Date	Stocks Affected	Avg Return	Event
Mar 23 2020	48	-12.94%	COVID-19 crash bottom
Mar 12 2020	48	-8.61%	COVID-19 initial crash
Aug 24 2015	46	-6.48%	China-triggered global selloff
Mar 16 2020	45	-7.23%	COVID-19 crash (continued)
May 18 2009	44	+14.90%	Post-election UPA rally
Oct 24 2008	43	-9.85%	Global Financial Crisis
Jan 21 2008	43	-8.66%	GFC — pre-Lehman crash
Apr 7 2020	42	+8.70%	COVID relief bounce

EDA Key Takeaway: Fat tails and volatility clustering are pervasive across all NIFTY-50 stocks. Normality assumptions severely underestimate crash risk. The average pairwise correlation of 0.257 confirms meaningful diversification potential, while same-sector stocks (e.g. TCS–INFY ~0.6, HDFCBANK–SBIN ~0.7) move in tandem — making cross-sector allocation essential.

2. Feature Engineering

Over 30 technical features are computed from raw OHLCV data, spanning four categories to give models a comprehensive view of market conditions.

Category	Features Computed
Trend	SMA (5/10/20/50/200-day), EMA (12/26-day), MACD + signal + histogram, price position vs. moving averages
Momentum	RSI-14 (Wilder smoothing), Stochastic %K & %D, Rate of Change (5/10/20/60-day), N-day price momentum
Volatility	Bollinger Bands (20-day, 2σ), ATR-14, Historical vol (5/10/20/60-day rolling), Parkinson estimator
Volume	Volume moving averages & ratios, On-Balance Volume (OBV), Volume-Price Trend (VPT)

Selection rationale: Features were chosen for their established predictive power in technical analysis literature and their ability to capture trend, reversal, risk, and liquidity signals simultaneously. Volume-based and momentum features consistently ranked highest in XGBoost feature importance scores, often outperforming simple price-derived inputs.

3. Methodology & Model Architecture

3.1 Dual-Model Prediction Engine

Two complementary machine learning models are trained for each stock, targeting future N-day returns (configurable: 7–90 days). An 80/20 time-series-aware train/test split is used — no data leakage.

Aspect	XGBoost Regressor	LSTM Neural Network
Architecture	Gradient-boosted trees	2-layer LSTM (64 → 32 units)
Key params	200 estimators, depth=7 lr=0.05, 80% subsample	Dropout=0.2, 60-day lookback Adam optimiser + early stopping
Input	30+ engineered features	Raw scaled OHLCV sequence
Scaling	StandardScaler	MinMaxScaler
Strength	Feature importance Short-horizon accuracy	Sequential patterns Long-horizon context
Directional acc.	55–65%	53–61%

3.2 Evaluation Metrics

Models are evaluated on MAE (Mean Absolute Error), RMSE, R^2 score, and directional accuracy — the percentage of correct up/down predictions. Directional accuracy of 55–65% is meaningful above the 50% random baseline and provides actionable signal for portfolio rebalancing decisions. XGBoost generally dominates on shorter horizons; LSTM gains competitiveness as the lookback window grows.

3.3 Limitations

- Models trained on historical patterns — market regime changes can degrade accuracy.
- Predictions are most reliable for direction rather than exact return magnitude.
- Ensemble combination (XGBoost + LSTM) reduces variance and is used in production.
- No sentiment, fundamental, or macro-economic data is incorporated at this stage.

4. Portfolio Construction Logic

Portfolio construction is grounded in Modern Portfolio Theory (Markowitz, 1952). The optimiser maximises risk-adjusted returns subject to investor-specific constraints, using SciPy's SLSQP (Sequential Least Squares QP) algorithm.

4.1 Optimisation Framework

Three investment profiles are supported, each with a distinct objective function:

Profile	Objective	Target Return	Max Volatility	Horizon	Rebalancing
Conservative	Minimise variance	8–12%	15%	3–5 years	Quarterly
Balanced	Max return @ 20% vol	12–18%	25%	2–4 years	Quarterly
Aggressive	Maximise Sharpe ratio	18%+	40%	1–3 years	Monthly

4.2 Constraints & Diversification

- Fully invested: portfolio weights sum to exactly 100%.
- Long-only: no short positions allowed.
- Concentration cap: maximum 35% single-stock exposure.
- Minimum position threshold: 0.5% to exclude negligible allocations.
- Sector diversification: implicitly enforced by the optimiser, which naturally allocates to uncorrelated assets across sectors.

4.3 Efficient Frontier

The Efficient Frontier is computed by solving for optimal portfolios at 50 evenly spaced target volatility levels between the global minimum variance and maximum Sharpe portfolios. This quantifies the full risk-return tradeoff and enables investors to select a point matching their exact risk tolerance. Sector-wise allocation breakdowns are reported for each portfolio to ensure transparency.

Portfolio Insight: The average pairwise correlation of 0.257 among NIFTY-50 top stocks implies significant diversification potential. A cross-sector balanced portfolio can achieve substantially higher Sharpe ratios than any single-sector concentrated bet — particularly avoiding the high-vol Financial Services cluster (avg vol 45.5%).

5. Risk Assessment Methodology

5.1 Ten-Metric Risk Framework

Each stock and portfolio is evaluated across 10 complementary risk metrics, giving investors a multi-dimensional view of downside exposure:

#	Metric	Formula / Method	Insight
1	Annualised Volatility	$\text{Std}(\text{daily ret}) \times \sqrt{252}$	Overall risk level
2	Sharpe Ratio	$(\text{Ret} - \text{Rf}) / \text{Vol}$	Risk-adj. return
3	Sortino Ratio	$(\text{Ret} - \text{Rf}) / \text{Downside vol}$	Penalises only losses
4	Maximum Drawdown	Max peak-to-trough %	Worst loss scenario
5	VaR (95%)	Historical 5th percentile	Daily loss threshold

6	CVaR (95%)	Mean of losses beyond VaR	Expected tail loss
7	Calmar Ratio	Ann. return / Max DD	Drawdown efficiency
8	Omega Ratio	P-weighted gains / losses	Full distribution view
9	Return Skewness	Third standardised moment	Asymmetric risk
10	Excess Kurtosis	Fourth moment - 3	Tail / fat-tail risk

5.2 Risk Classification & Rolling Analysis

Category	Volatility Threshold	Max Drawdown Threshold
Low Risk	< 18%	< 30%
Moderate Risk	< 30%	< 50%
High Risk	< 45%	—
Very High Risk	≥ 45%	—

Rolling volatility and Value at Risk are computed over moving windows to track how risk evolves through time. This is particularly valuable for identifying market stress regimes — e.g. RELIANCE's annualised volatility spiked well above 60% during the COVID-19 crash of March 2020, versus its long-run average of 38.2%.

6. Explainability Techniques (XAI)

Explainability is a first-class design requirement — every recommendation must be accompanied by a transparent, human-readable rationale. The framework operates at four levels:

XAI Layer	Technique	User-Facing Output
Feature Importance	XGBoost built-in gain scores	Top-5 predictive features with percentage contribution
Natural Language Generation	Template-based NLG on quantitative signals	Plain-English prediction summary with confidence
Portfolio Rationale	Per-stock metric comparison to portfolio mean	Why each stock was selected or excluded
Risk Interpretation	Threshold-based contextual commentary	Risk metrics compared to benchmarks and thresholds

6.1 Explanation Structure

Every model output follows a four-part explanation format: **(1) Summary** — a single-line takeaway; **(2) Key Factors** — the top-5 influencing features with importance scores; **(3) Context** — comparison to historical benchmarks and thresholds; **(4) Caveats** — explicit statement of model limitations and confidence levels. This structure ensures that even non-technical users can evaluate the quality of a recommendation before acting on it.

7. Key Insights

7.1 Market Behaviour

Tail risk is real: Return distributions across all 49 NIFTY-50 stocks exhibit excess kurtosis, meaning extreme daily moves occur far more often than Gaussian models predict. Standard VaR models underestimate crash risk by a significant margin.

Volatility clusters: High-volatility regimes (e.g. 2008 GFC, March 2020) persist for weeks or months. GARCH-style rolling risk windows capture this better than static metrics.

Sector diversification pays: Average same-sector correlation exceeds 0.60 (e.g. TCS–INFY, HDFCBANK–SBIN), while cross-sector pairs average 0.20–0.35. Multi-sector allocation significantly improves the Sharpe ratio of any portfolio.

Event detection works: 281 market-wide event days were automatically identified — each corresponding to a known macro shock — demonstrating that statistical anomaly detection provides reliable early-warning signals.

7.2 Model & Portfolio Insights

Volume & momentum lead: Volume-based features and momentum indicators consistently rank highest in XGBoost feature importance, outperforming simple price-level features. This validates using OBV and VPT alongside RSI and MACD.

Ensemble beats singles: Combining XGBoost and LSTM via a simple average reduces prediction variance and improves directional accuracy by 2–4 percentage points over either model alone.

Horizon matters: The 30-day prediction horizon delivers the best balance of reliability and actionability. Accuracy degrades noticeably beyond 60 days.

Cement sector outperformed: Over 21 years, Cement & Cement Products generated 15,179% average total return — though with 37.3% annualised volatility. Risk-adjusted, Financial Services and Consumer Goods offered superior Sharpe ratios.

8. Conclusion

This project delivers a comprehensive AI-powered investment intelligence platform for the NIFTY-50 universe, combining machine learning prediction, Markowitz portfolio optimisation, a 10-metric risk framework, statistical anomaly detection, and a structured explainability layer — all surfaced through an interactive Streamlit dashboard.

Component	Status	Key Capability
Stock Prediction Engine	Complete	XGBoost + LSTM; 55–65% directional accuracy
Portfolio Optimiser	Complete	Markowitz MPT; 3 risk profiles; SLSQP solver
Risk Analytics	Complete	10 metrics incl. CVaR, Sortino, Calmar, Omega
Anomaly Detection	Complete	281 market events detected; 3-method consensus
Explainable AI	Complete	Feature importance + NLG explanations
Interactive Dashboard	Complete	Streamlit UI; fully reproducible codebase

Future Work

- Reinforcement Learning for dynamic portfolio rebalancing.
- Bayesian methods for uncertainty quantification in predictions.
- Transformer models (e.g. Temporal Fusion Transformer) for long-range sequence modelling.
- Multi-asset integration: bonds, gold, and REITs alongside equities.
- Real-time data pipeline for live NIFTY-50 monitoring.

9. References

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